

PH 530 : Cosmology-I

Course Code : PH 530

Course Name : Cosmology-I

L-T-P-C : 3-0-0-3

Intended for : UG/PG/PhD/PhD elective

Prerequisite :

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction:** What is Cosmology? A brief history of the Universe a Cosmological erasination, reheating, recombination,etc. Time and length scales involved in Cosmology What is our beloved universe made of? (1 Hour)
- **Homogeneous Universe:** The Cosmological Principle Geometry of the Universe a The FLRW Metric Kinematics a The Hubble Law a Geodesics a Redshift Distances a Luminosity a Angular Diameter Dynamics a Energy Budget of the Universe a Friedmann Equations Some Exact solutions Our Universe. (14 hours)
- **Hot Big Bang:** Thermal Equilibrium a Some aspects of Statistical Mechanics a Primordial Plasma a Entropy and Expansion History a Cosmic Microwave Background a Cosmic Background Beyond Equilibrium a The Boltzmann Equation a Dark Matter Freeze Out a Big Bang Nucleosynthesis More on recombination. (15 hours)
- **Cosmological Inflationary Theory:** Problems faced by the Big Bang Theory a The Horizon Problem a The Flatness Problem a Superhorizon Correlations Before the Hot Big Bang The Physics of Inflation a How inflation solves these problems? a Slow Roll Inflation. (12 hours)

Textbooks:

1. Daniel Baumann, **Cosmology**, Cambridge University Press, 2021
2. Scott Dodelson & Fabian Schmidt, **Modern Cosmology**, Academic Press, Elsevier, 2021.

References:

1. Steven Weinberg, **Cosmology**, Cambridge University Press, 2008
2. Valery Rubakov and Dmitry Gorbunov, **Introduction to the theory of Early Universe**, Volume 1&2, World Scientific, 2011
3. Luca Amendola, **Cosmology Lecture Notes**, University of Heidelberg, 2023
4. Viatcheslav Mukhanov, **Physical Foundations of Cosmology**, Cambridge University Press, 2005.
5. Andrew Liddle, David Lyth, **Cosmological Inflation and Large Scale Structure**, Cambridge University Press, 2000.